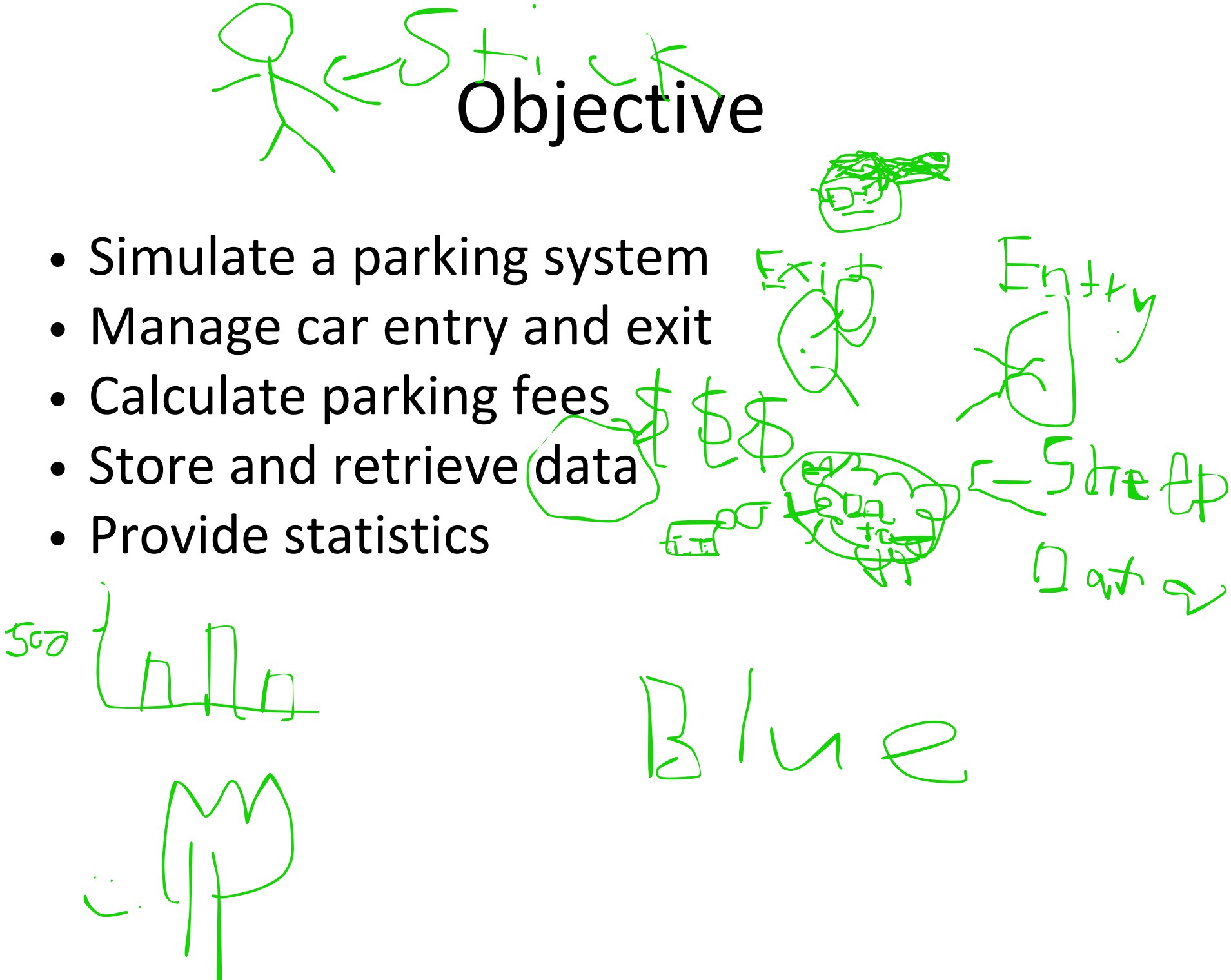
Parking Management System

- A C program that manages a parking lot, tracks cars, calculates fees, and stores data using files.

Objective

- Simulate a parking system
- Manage car entry and exit
- Calculate parking fees
- Store and retrieve data
- Provide statistics



Data Structure

- Each car has an ID, license plate, entry time, and exit time.
- Used to store information about each parked car.

Main Features

- Add car
- Remove car
- Display cars
- Show statistics
- Save and load data
- Backup and restore
- Exit system

```
========== PARKING SYSTEM ==========
1. Car enters parking
2. Display parked cars
3. Car leaves parking
4. Number of parked cars
5. Statistics
6. Create backup save
7. Restore from backup
8. Reset current save
0. Exit
Choose:
```

Adding a Car

- User enters car ID, plate, and entry time.
- System checks validity and stores the car.
- Car is added to the parking list.

```
Choose: 1
Enter car ID: 11307
Enter license plate: Example_plate
Enter entry time (hour): 8
Car parked successfully.
```

Car Leaving

- User enters car ID and exit time.
- System calculates parking duration and fee.
- Updates total revenue and removes car.

```
Choose: 3
Enter car ID to leave: 11307
Enter exit time (hour): 15
Parking fee: 35000
Car with ID 11307 has left the parking.
```

Displaying Cars

- Shows all currently parked cars.
- Includes ID, plate number, and time information.

ID	Plate	EntryTime	ExitTime
11307	Example_plate	8	-1
12345	Example_plate_2	7	-1
13245	Example_plate_3	1	-1
14235	Example_plate_4	9	-1

After leaving

ID	Plate	EntryTime	ExitTime
12345	Example_plate_2	7	-1
13245	Example_plate_3	1	-1
14235	Example_plate_4	9	-1

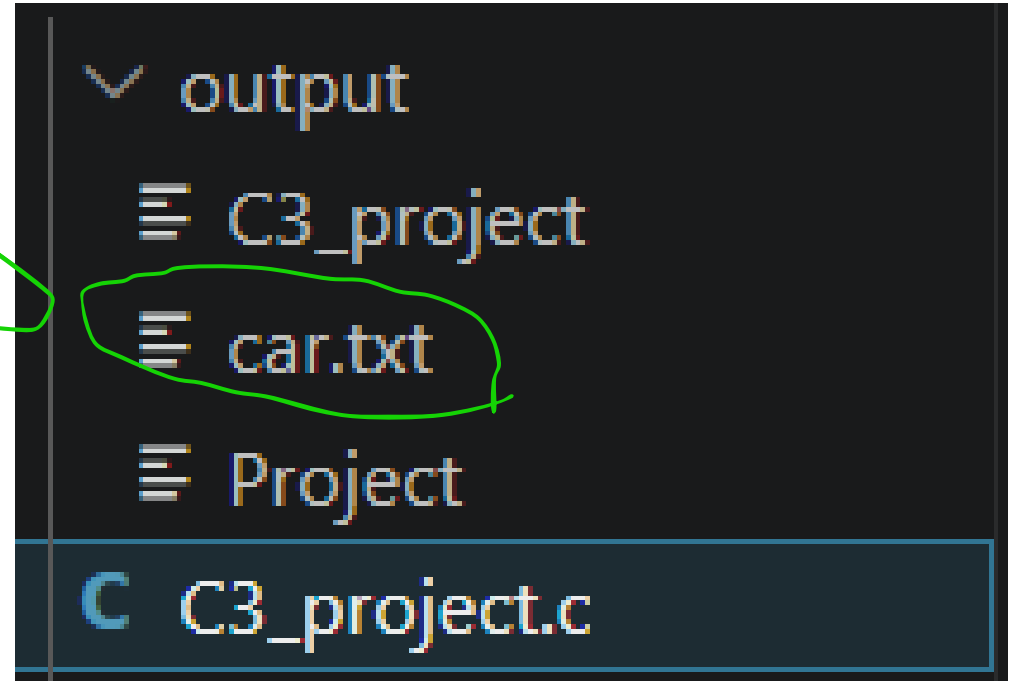
Statistics

- Displays total cars parked, total revenue, and total cars served.
- Identifies the car with the longest parking duration.

```
PARKING STATISTICS
Total cars parked: 3
Total parking fees collected: 35000
Total cars served: 1
```

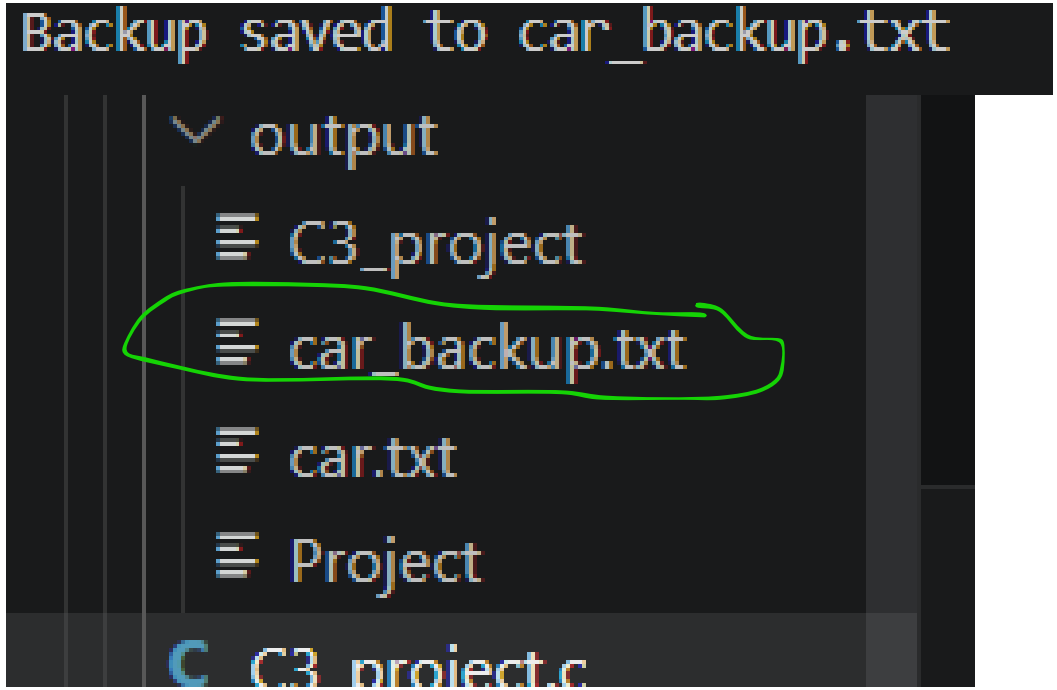
Saving n Loading Data

- All data is saved to a file.
- Ensures information is not lost after program exits.
- Reads saved file when program starts.
- Restores previous parking data and statistics.
- Save



Backup System

- Creates a backup copy of saved data.
- Protects against data loss.
- Restores data from backup file.
- Useful if main file is corrupted.



Reset System

- Clears all stored data.
- Resets counters and saves empty state.
- Necessary for update and fixes of the code

Program Flow

- Load saved data
- Display menu
- User selects option
- Perform action
- Save data
- Repeat until exit